AssistVision: A Turntable–Load-Cell Rig and a Controlled Benchmark of Monocular versus Multi-View Food Volume Estimation

[Author names]
[Affiliation]

***Abstract*—Image-based portion estimation needs metric volume, and a single image does not contain metric scale. We build a low-cost capture rig (ESP32-S3 camera, stepper turntable, HX711 load cell, printed ArUco mat) that returns N calibrated views together with the true mass of the plate, and we ask, with controlled experiments rather than anecdotes, when a single-view depth network can replace multi-view reconstruction. On the 507-dish Nutrition5k depth test split we benchmark [n models] monocular depth models under three explicitly separated calibration protocols (raw, deployable anchor, oracle upper bound) and report bootstrap confidence intervals and paired tests. On [n synth] synthetic plates with exact mesh volume we compare dense multi-view stereo with monocular depth and ablate view count, sensor resolution, JPEG quality, marker-scale error and plate material. [abstract numbers]**

I. Introduction

Dietary assessment by photograph is attractive because it removes manual logging, but portion size—grams—is the quantity nutrition depends on and the one images recover worst. Perspective projection discards absolute scale: a small plate near the camera and a large plate far away produce the same pixels. Any system must inject metric scale from outside the image: from a reference object, from a depth sensor, from camera motion with known geometry, or from a learned prior. This paper measures how much each of these choices costs.

Our contributions are:

1) An open hardware capture rig (Sec. IV) that produces multi-view images, metric scale and gravimetric ground truth in one button press, with firmware, host software, wiring and a bench-test protocol.
2) A benchmark of monocular depth models for food on real RGB-D data (Nutrition5k) that separates deployable calibration from oracle calibration, a distinction prior food-volume work often blurs.
3) A controlled synthetic study with exact volumes that isolates the factors the rig designer actually controls: views, resolution, compression, marker accuracy and serveware material.
4) A documented record of design decisions and rejected alternatives (Sec. III), each tied to a measurement or a stated assumption.

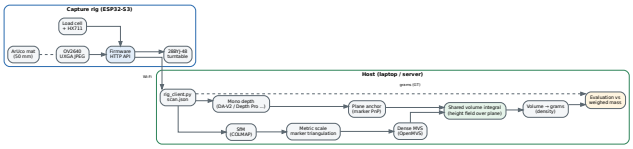


Fig. 1. System architecture. The rig exposes an HTTP API; the host drives the scan loop, so dropped frames are retried rather than corrupting a scan.

II. Related Work

Food volume. Nutrition5k [1] provides overhead RGB-D video and weighed dishes; its depth-based volume baseline uses a constant per-pixel footprint at the capture-plane distance. Multi-view food reconstruction has used structure-from-motion with reference cards and dense stereo [2], [3]. **Monocular depth.** Relative depth models (MiDaS [4], Depth Anything [5], [6]) predict depth up to an unknown affine transform of inverse depth; metric models (ZoeDepth [7], Depth Pro [8], metric-finetuned Depth Anything V2) predict absolute depth from priors learned on room- and street-scale scenes. **Reconstruction.** We use COLMAP [9] for SfM and OpenMVS for dense MVS.

III. Design Decisions and Rejected Alternatives

Table I summarises each decision; the evidence column points to the experiment that supports it. Decisions marked *assumption* rest on datasheets or cost quotes, not on our measurements.

IV. Hardware

The rig (Figs. 1–3) uses an ESP32-S3 camera board (OV2640, up to 1600 × 1200), a 28BYJ-48 stepper through a ULN2003 driver (4076 half-steps per revolution, 0.088° per step), and a 5 kg bar load cell read by an HX711 at 10 Hz. The firmware exposes `/capture`, `/rotate`, `/weight`, `/tare`, `/calibrate` and camera-setting endpoints. Exposure, gain and white balance are locked after convergence so that all views of a scan share photometry, which MVS assumes. Stepper coils are released while weighing to avoid heating and vibration of the load cell. Tare and the counts-per-gram

TABLE I
DESIGN DECISIONS, ALTERNATIVES CONSIDERED, AND THE EVIDENCE BEHIND EACH CHOICE.

Decision	Chosen	Rejected alternatives	Why / evidence
Scale source	Printed ArUco mat, multi-view triangulation	Credit-card reference in one image; metric monocular depth; depth sensor	Volume scales with the *cube* of length error (Sec. V-B); metric networks are off by large factors at plate distance (Table II); a D435-class sensor costs $\sim$10$\times$ the camera (assumption).
Geometry	Dense MVS (COLMAP + OpenMVS)	Sparse SfM points only; single-view depth	Sparse points under-cover low-texture food (Sec. VI-C); single view cannot see the far side of a mound even with perfect depth.
Capture motion	Fixed camera, motorised turntable	Hand-held phone orbit; multi-camera ring	Repeatable baselines and known angles; one camera keeps cost and synchronisation trivial; the angle prior is checked against SfM.
Camera tilt	45° elevation	Straight overhead (Nutrition5k)	Overhead view makes the support plane fronto-parallel: the relative-depth anchor fit becomes rank-deficient (condition number 10^{14}, Sec. V-C).
Controller	ESP32-S3 + OV2640	Raspberry Pi + camera; smartphone app	Pi adds cost, boot time and an OS to maintain; phones vary per user and cannot drive the turntable or scale. Processing is host-side, so the MCU only needs camera, Wi-Fi and GPIO (assumption: cost table in hardware README).
Ground truth	5 kg bar load cell + HX711 under the plate	External kitchen scale; no mass at all	In-rig weighing removes a manual step and lets every scan carry its own reference; tare-before-serve gives food-only mass.
Volume integral	Height field over the fitted support plane	Constant per-pixel footprint (Nutrition5k); voxel carving; mesh + watertight closure	Exact on tilted planes in our unit tests (0.09 % error) where the constant-footprint formula is not; shared across all methods so comparisons measure geometry, not bookkeeping.
Mono calibration	Report raw / anchor / oracle separately	Report only oracle-aligned numbers	Oracle alignment uses the ground truth being evaluated; it is an upper bound, not a deployable result.

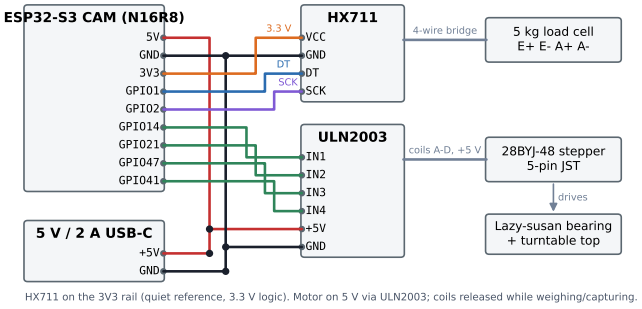


Fig. 2. Wiring of the ESP32-S3, HX711 load-cell amplifier and ULN2003 stepper driver.

calibration persist in flash. The firmware compiles for the ESP32-S3 target with the Arduino core 3.3.12; a fake-firmware test exercises every client endpoint, retry on a dropped frame, and calibration arithmetic.

Physical validation (pending). Load-cell linearity, repeatability and drift, turntable angular error, scan duration and transfer latency are specified as scripted bench tests (`rig_client.py bench-*`) whose results are [to be measured on the assembled rig]. We report no physical number we have not measured.

V. METHOD

A. Shared volume integral

All methods produce a 3-D point set P in metres and a support plane $(\mathbf{n}, d)$ oriented towards the camera. Each point gets a height $h = \mathbf{n}^\top \mathbf{p} + d$ and plane coordinates in an orthonormal basis; points are binned into square cells of side c and each cell keeps the median height. Empty cells inside the footprint are filled by nearest neighbour, negative heights are clipped, and

$$V = c^2 \sum_{\text{cells}} \max(h, 0). \tag{1}$$

Because monocular, multi-view and sensor depth all share this integral, differences in volume reflect differences in geometry, not in post-processing.

B. Metric scale from the marker mat

Marker corners are detected in every view and triangulated with the SfM cameras; the scale s is the ratio of the known side to the median triangulated side. A relative error ϵ in s produces

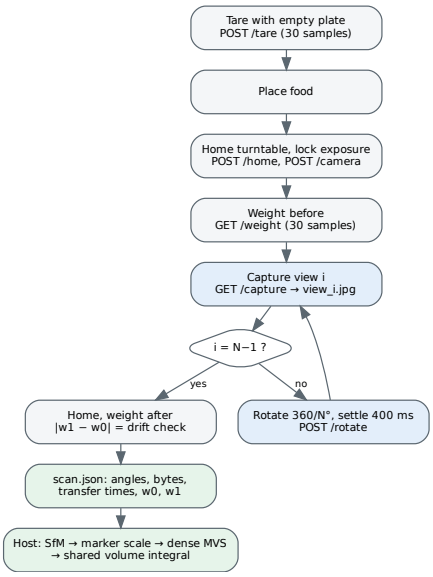


Fig. 3. Scan protocol implemented by `rig_client.py scan`. Weight is read before and after the turntable moves; their difference is a per-scan drift/vibration check.

[figure]

Fig. 4. Per-dish absolute volume error versus sensor depth, by model and protocol (Nutrition5k).

a volume error of $(1+\epsilon)^3 - 1 \approx 3\epsilon$, so a 1 % scale error costs about 3 % in volume; Sec. VI-C measures this directly.

C. Monocular protocols

Raw: the model's metric output is used as-is. **Anchor**: the prediction is calibrated using only the support plane (the table in Nutrition5k, the marker mat in our rig), never the food; metric outputs get one scale, relative outputs an affine map of inverse depth. **Oracle**: the same maps fitted to ground-truth depth over the whole dish region—an upper bound that is not available at deployment. For relative models we record the condition number of the anchor fit; a fronto-parallel plane has nearly constant inverse depth, so scale and shift are not separately identifiable.

VI. Experiments

A. E1: Monocular depth on Nutrition5k

[E1 findings]

B. E2: Sensor-referenced volume and weighed mass

Nutrition5k has no food-volume label, so the volume target in E1/E2 is the volume integrated from RealSense depth over the dish region with the same height-field integral (Eq. 1) applied to each model's prediction. The error therefore isolates the depth model: a model with zero error here reproduces the sensor, whatever the sensor's own error on food is. [E2 table]

[figure]

Fig. 5. Accuracy–latency trade-off (anchor protocol; CPU latency on the development VM).

C. E3: Multi-view versus monocular on exact synthetic volumes

Scenes. Twenty-two Blender (Cycles) scenes with watertight food meshes whose volume is computed exactly from the mesh: eight food configurations (single rice mound, rice + dal, two idli, curry heap, lumpy rice, a three-item thali, a 40 mL small portion and a tall mound) on ceramic and polished steel plates, four textureless variants and two on banana leaf. Each scene is rendered at 1600×1200, 60° horizontal FoV, 45° elevation and 0.40 m distance, with the plate rotating on a turntable inside a printed 50 mm ArUco mat, i.e. the rig geometry. View 0 also stores a metric depth map, which gives the "perfect single-view depth" row: the error a monocular method would have even if its depth were exact, because it cannot see behind the food.

Methods. (i) Dense MVS: COLMAP SfM, marker-triangulated scale, OpenMVS densification, plate-noise-calibrated footprint threshold, shared integral. (ii) Sparse SfM points only, same integral. (iii) Monocular depth on view 0 under the anchor and oracle protocols, with the plane taken from the marker-mat pose.

[E3 findings]

D. E4: Ablations

All ablations use the dense MVS pipeline and change one factor at a time: number of views (6, 8, 12, 16, 24, evenly spaced), image resolution and JPEG quality (the OV2640 range), a deliberate error in the printed marker size (± 1, $\pm 2\%$), plate material, and three repeated runs of the identical input to measure the run-to-run spread of SfM/MVS, which are not bit-deterministic. Reconstruction failures (SfM registering fewer than half the views, or no marker triangulated) are counted, not dropped. [E4 findings]

VII. Limitations

Nutrition5k has no volume ground truth. The dataset provides overhead RealSense depth and weighed mass, not food volume. We therefore score monocular models against the volume integrated from sensor depth over the same dish region (food plus plate rim above the support plane). This measures how well a model reproduces a depth sensor, not how well it measures food. Our attempt to separate food from plate with a second plate-floor plane fit gave a food volume that barely tracks weighed mass (Spearman $\rho = 0.24$, Pearson $r = -0.01$, $n = 502$), so we do not use it as a target. Absolute food-volume accuracy is reported only on synthetic scenes with exact mesh volume.

Synthetic-to-real gap. Blender scenes have perfect camera intrinsics, controlled lighting and procedural textures. Real OV2640 frames add rolling shutter, auto-exposure drift, lens

TABLE II

E1 (NUTRITION5K, n DISHES PER MODEL AS SHOWN). DEPTH ERROR OVER THE DISH REGION AND VOLUME ERROR RELATIVE TO SENSOR-DEPTH VOLUME, PER MODEL AND CALIBRATION PROTOCOL. BRACKETS: 95% BOOTSTRAP CI (10 000 RESAMPLES). LATENCY IS CPU TIME PER IMAGE ON THE DEVELOPMENT VM.

[E1 model table]

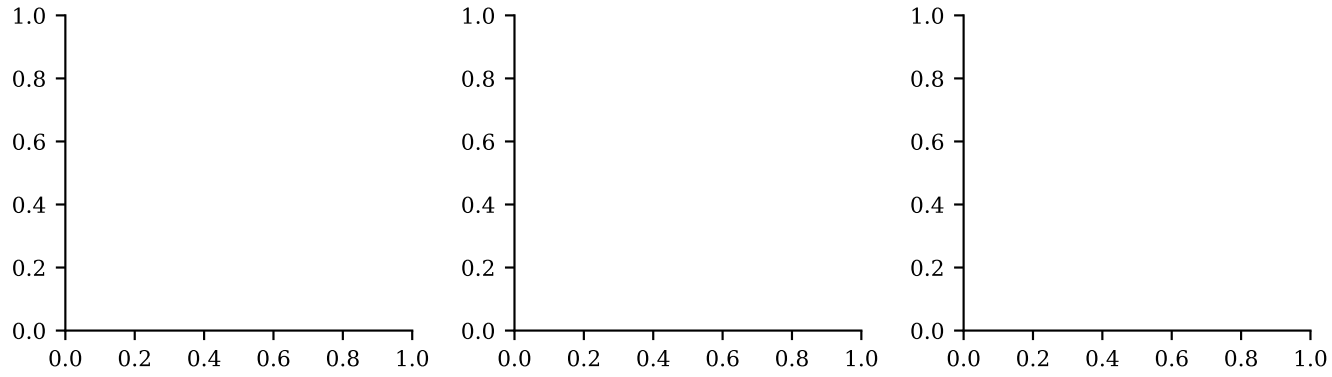


Fig. 6. E4 ablations (synthetic). Left: views per scan. Middle: resolution × JPEG quality. Right: marker-size error against the cube law.

TABLE III

E3 (SYNTHETIC, EXACT MESH VOLUME). ABSOLUTE VOLUME ERROR, MEDIAN AND 95% BOOTSTRAP CI OVER SCENES; BIAS IS THE MEAN SIGNED ERROR. TIME IS VM CPU SECONDS PER SCENE.

Method	Plate	n ok	—err— % median [95% CI]	bias %	<10%	s
MVS dense (24 views)	all	2/2	0.8 [0.2, 1.4]	-0.8	2	132
MVS dense (24 views)	ceramic	1/1	1.4	-1.4	1	132
MVS dense (24 views)	steel	1/1	0.2	-0.2	1	132

distortion and JPEG blocking on low-texture food. The resolution and JPEG ablations bound part of this gap but not all of it.

No physical measurements yet. Load-cell linearity, repeatability and drift, turntable angular error, scan duration and Wi-Fi transfer latency have scripts and empty tables (Sec. IV); none of the numbers in this paper come from the physical rig. Timings are CPU timings on the development VM, not on the ESP32-S3.

Volume is not mass. Converting volume to grams needs a per-food density. We do not estimate density; the load cell measures mass directly, which is the reason it is part of the rig.

VIII. CONCLUSION

[conclusion results]

REPRODUCIBILITY

All code, scene specifications, per-dish result files and the scripts that generate every table and figure are in the repository. Evidence categories are labelled throughout: *Nutrition5k* (public real RGB-D data), *synthetic* (Blender/Cycles renders with exact mesh volume), *VM* (timings on an 8-core CPU-only cloud VM, not on the ESP32), and *physical* (pending rig measurements).

REFERENCES

[1] Q. Thames, A. Karpur, W. Norris, F. Xia, L. Panait, T. Weyand, and J. Sim, "Nutrition5k: Towards automatic nutritional understanding of generic food," in *CVPR*, 2021, pp. 8903–8911.

[2] J. Dehais, M. Anthimopoulos, S. Shevchik, and S. Mougiakakou, "Two-view 3d reconstruction for food volume estimation," *IEEE Trans. Multimedia*, vol. 19, no. 5, pp. 1090–1099, 2017.

[3] F. P. W. Lo, Y. Sun, J. Qiu, and B. Lo, "Image-based food classification and volume estimation for dietary assessment: A review," *IEEE J. Biomed. Health Inform.*, vol. 24, no. 7, pp. 1926–1939, 2020.

[4] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, and V. Koltun, "Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer," *IEEE TPAMI*, vol. 44, no. 3, pp. 1623–1637, 2022.

[5] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth anything: Unleashing the power of large-scale unlabeled data," in *CVPR*, 2024.

[6] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything v2," in *NeurIPS*, 2024.

[7] S. F. Bhat, R. Birkl, D. Wofk, P. Wonka, and M. Müller, "Zoedepth: Zero-shot transfer by combining relative and metric depth," *arXiv:2302.12288*, 2023.

[8] A. Bochkovskii, A. Delaunoy, H. Germain, M. Santos, Y. Zhou, S. R. Richter, and V. Koltun, "Depth pro: Sharp monocular metric depth in less than a second," in *ICLR*, 2025.

[9] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *CVPR*, 2016.